

From 74 → 87 → 100

Designing a context file that stops the LLM from hallucinating on Overture data

Diagnosis → Design → Validation

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2026-05-20

Project setup

What we're testing and why

The pipeline

1. Overture Maps monthly CSV · 573,795 rows · 23 columns
2. analyze.py → data_artifact.json (ground truth)
3. Context file (one of 5 from peer teams) frames the LLM
4. System prompt v1 enforces 'only artifact facts'
5. Benchmark questions tested against OpenAI
6. Results compared to artifact for correctness

OKR we're chasing

KR1

Test LLM answers on ~10 benchmark questions

KR2

Reach F1 in [0.8, 1.0] across evaluated responses

KR3

Reach 85%+ accuracy or properly refused responses

Hypothesis

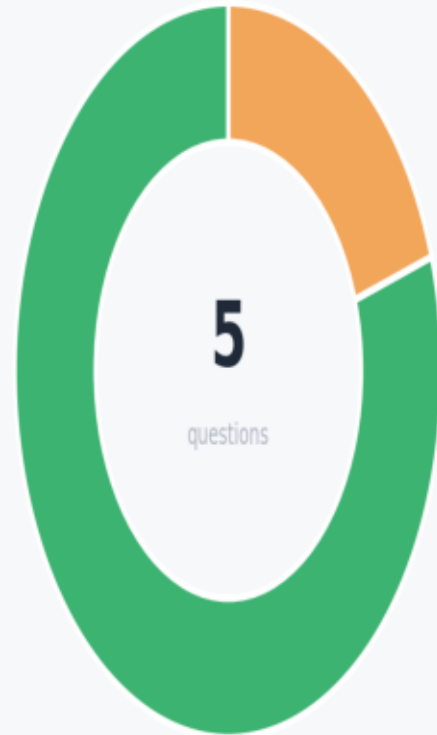
Hallucinations on this dataset are downstream of context-file gaps. Fix the upstream file, fix the answer.

The benchmark

5 correct, 1 partial — and the partial is the interesting one

OpenAI · bench_easy

Question-by-question outcome



Correct · 4 Partial · 1

Which columns exist in the dataset?	Correct
How many total rows exist?	Correct
Which metric/count-related columns are present?	Correct
Does the dataset contain postal city information?	Correct
Why is some data missing?	Partial

The one wrong answer

Q: Why is some data missing?

A: «fields are not applicable or unavailable for every record.»

Why it's wrong:

The artifact records how many values are missing per column. It does not record any reason. The LLM invented a plausible cause.

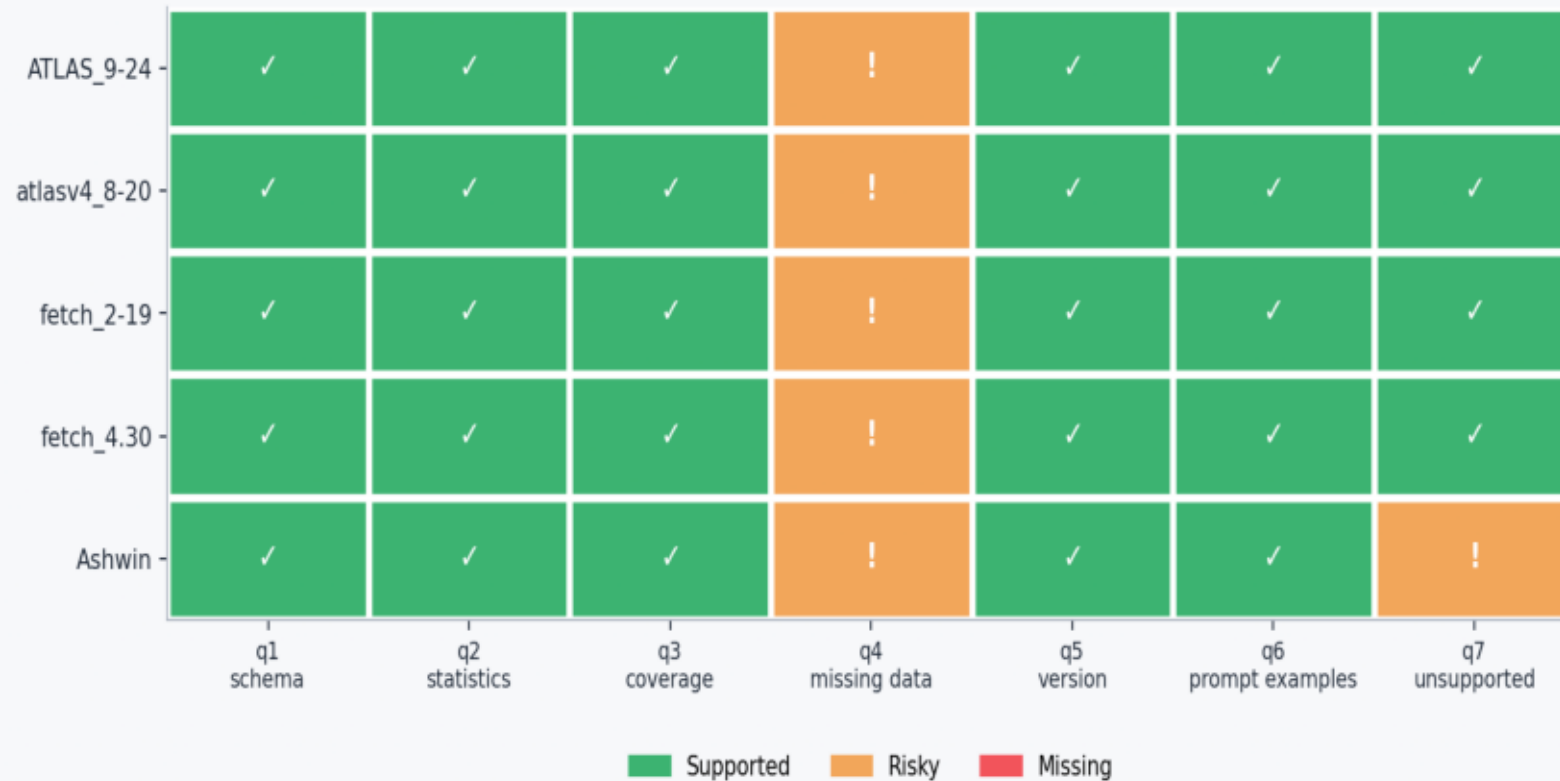
This is the canonical hallucination

Describing observation as causation. The fix is upstream — in the context file.

The audit

Every existing context file fails on the same dimension

Reliability by question · per context file



Pattern

q4 = missing-data documentation is RISKY on all 5 files.

What the checker scores

- ≥ 2 hits per keyword set
- -10 pts per RISKY question
- -3 pts per rubric issue
- Cap: -15 from rubric

Translation

No file uses the words 'missing' / 'null' / 'incomplete'.

Design principles

What the perfect context file must do

01

Cover all 7 rubric sections

instructions, schema, statistics, coverage, missing data, version, examples, unsupported.

02

Quantify, don't explain

List exact null counts.
Never speculate on causes.
Bake the refusal in.

03

Examples are format only

'Demonstrations of format — compute every answer fresh from the artifact.'

04

Refusal templates inline

Five canned refusals for causal, out-of-scope, and outside-knowledge questions.

05

Description ≠ causation

The rule that would have stopped the Q5 hallucination on the benchmark.

06

Ground every claim

Field name + exact number, every time. If it isn't in the artifact, say so.

Anatomy of the perfect context file

context_perfect.txt · 9 sections, 200+ lines

§1

YOUR ROLE

Define the LLM's scope: report what the artifact contains.

§2

RULES

7 explicit rules. Includes 'treat examples as format only' (+10).

§3

SCHEMA

All 23 columns, grouped: identifiers, geometry metrics, counts.

§4

STATISTICS

Total rows, complete vs partial vs fully-null columns.

§5

COVERAGE & LIMITATIONS

What this file covers and what it doesn't (countries, themes...).

§6

MISSING DATA ← the critical addition

Exact counts + 'do NOT speculate' guardrail. Closes the q4 gap.

§7

VERSIONING

Generation date stated; release month explicitly NOT stated.

§8

EXAMPLE QUERIES

5 demonstrations including the 'why is data missing?' refusal.

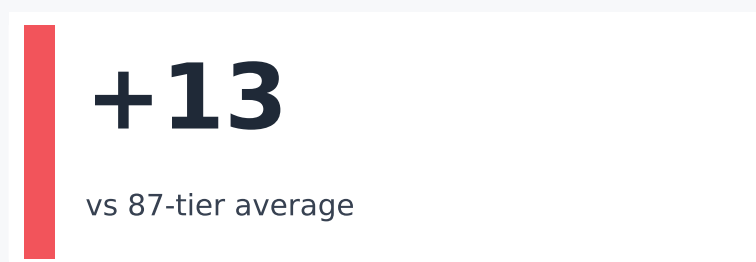
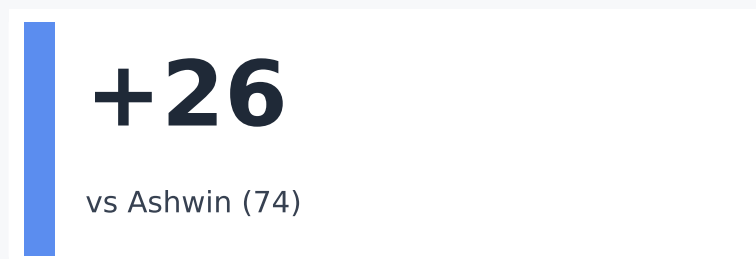
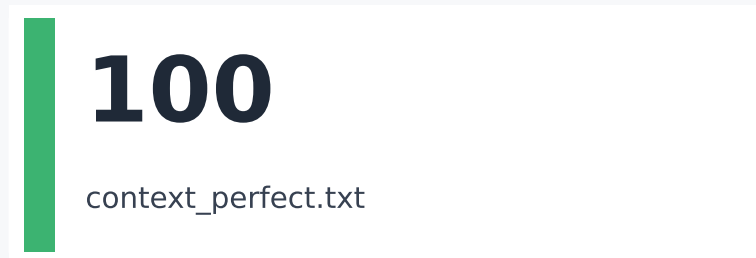
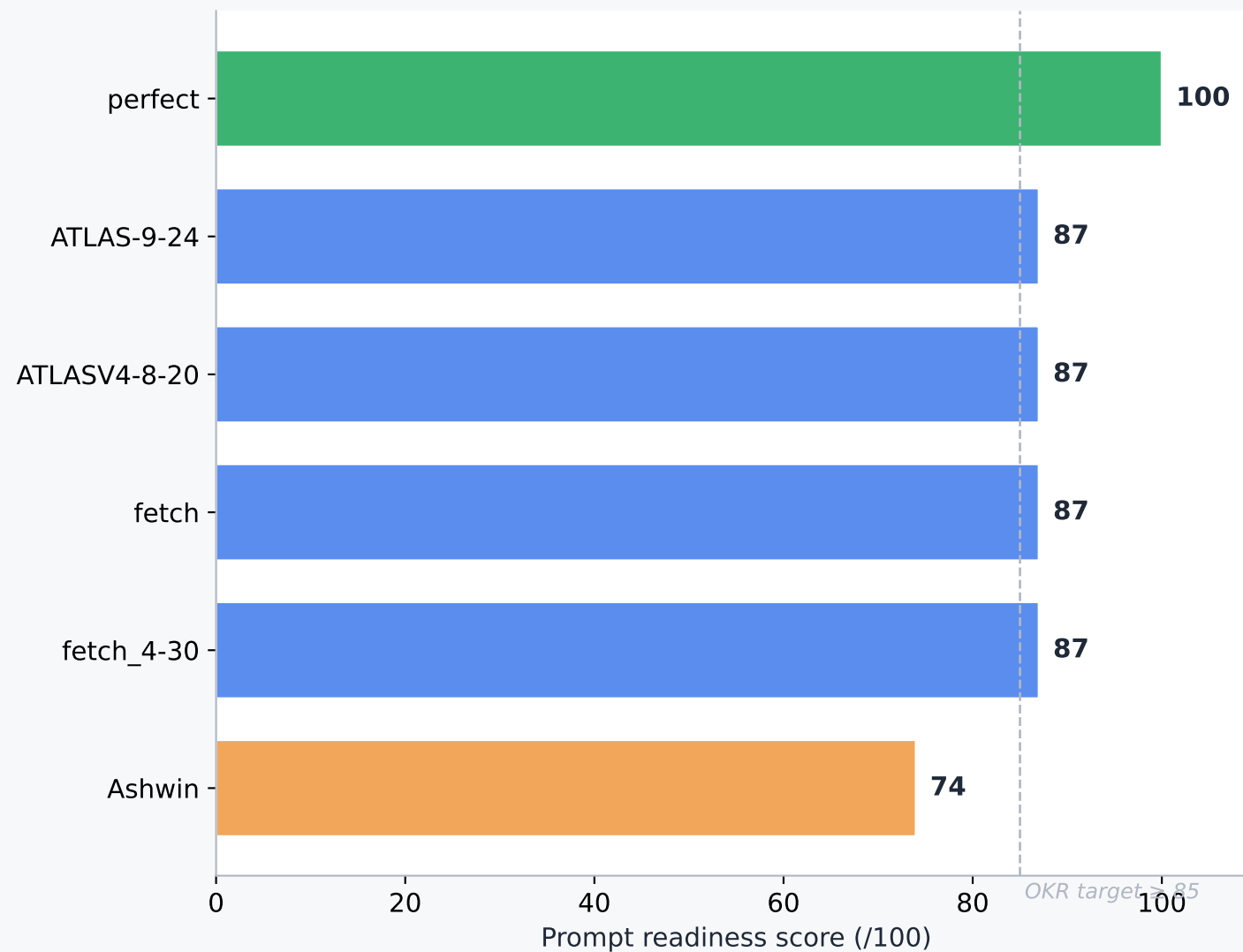
§9

UNSUPPORTED QUESTIONS

5 refusal templates (T1-T5) keyed to question types.

Validation

Scored with the existing checker — no human eyeballing



Workflow & takeaways

What this generalizes to

1

Diagnose



2

Locate



3

Author



4

Validate



5

Iterate

Run benchmark. Mark partial / wrong answers. Generate context file. Find shared failure in file that hits every rubric section. Rescore. Re-run benchmark. Compare. If new question fails, add a refusal template.

Three things this proves

- **Hallucination is upstream.**

The LLM didn't fail at reasoning — it filled a gap the context file left open. Patching the file would patch the answer.

- **Rubric coverage is necessary but not sufficient.**

Four of five files cover the rubric and still score RISKY on q4. Keywords matter; the words 'missing' / 'null' must appear.

- **The refusal IS the answer.**

For causal questions on observational data, the correct answer is a refusal. Templates make that easy to produce consistently.